White paper: should biomarker moderation be encoded as a latent class?

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A two-page summary of 'Latent-class and mixture-model formulations for biomarker-treatment interaction in N-of-1 trials' (pmsimstats team, 2026-06-13).

Goal of the paper

Aggregated N-of-1 trials exist to detect a biomarker-by-treatment interaction. Published practice encodes that interaction one way only: as a smooth continuous term, $\beta_{bm:D} \cdot B \cdot D$, in a single-component regression. Every participant is assumed to respond to some degree, and the biomarker is assumed to scale that response linearly.

There is a second, equally defensible encoding. Participants may belong to unobserved responder and non-responder classes with qualitatively distinct response curves, with the biomarker predicting class membership without determining it. This is a finite mixture, and the psychometric, econometric, and marketing-science literatures have developed mature machinery for it (latent class and profile analysis, growth mixture models, factor mixture models, regression mixtures and mixtures of experts) that the N-of-1 biostatistics literature has not engaged.

The paper's claim is that the choice between the two encodings is a substantive scientific question, not a parameterization convenience, and it pursues four aims:

1. Formalize the latent-class data-generating process for aggregated N-of-1 trials and derive when a single multivariate normal can stand in for it.
2. Characterize estimation and the identifiability conditions at trial-relevant sample sizes, $N \in [30, 150]$.
3. Benchmark, by simulation, whether a class-aware analysis beats the standard linear mixed model.
4. Re-analyze the Hendrickson et al. (2020) prazosin-PTSD data under the latent-class formulation.

Aims 1 and 2 are delivered in full. Aim 3 is delivered only as a 240-replicate pilot of the last of five planned studies. Aim 4 has not been carried out.

Why the encoding matters under carryover

The two encodings diverge precisely where N-of-1 trials are weakest. Carryover enters as a decaying drug indicator $D_{it} = \exp(-\lambda t_{sd})$. Under continuous heterogeneity the moderation enters as a product with D_{it}, so carryover attenuates the signal directly, which is the mechanism behind the 40-60% covariance-only power loss reported in the companion work. Under a latent-class DGP the class label is fixed across drug states, so some moderation information survives washout and is in principle recoverable from the full trajectory. The gap between the two power curves upper-bounds what a class-aware analysis could recover. The paper is explicit that this bound is asymptotic and can shrink or reverse at finite N.

The theory section also establishes when the MVN shortcut is safe. A two-component mixture is reproduced to second order by a single MVN with c_{bm}^2 equal to the product of the between-class variance fractions in B and in BR, so the achievable marginal correlation is bounded by the geometric mean of the two class separations. The approximation fails in three named regimes: bimodal response distributions, a near-step gating function (where a thresholded mean-moderation model is the better approximation), and class-varying covariance.

Results

The pilot. Study 5, 240 replicates per cell, $N = 70$, eight timepoints, a heterogeneous-random-slopes DGP with logistic gating in the standardized biomarker at gating slopes $\{0, 1.0, 1.5\}$, run on four cores in 558 s. Three tests per replicate: the linear-mixed `bm:Dbc` Wald, an unconditional Wald on the `lcmm::hlme` gating coefficient, and

a BIC-conditional gating procedure that tests the gating coefficient only if BIC prefers $ng = 2$. Monte Carlo standard error on power is about 0.030; `lcmm` convergence was 92.9% at the null and 98.3-98.8% elsewhere.

The unconditional class-aware test is badly anti-conservative. Type I error 0.197 (95% CI [0.145, 0.250]) against nominal 0.05, more than five MCSE of separation. The diagnosis is a finite-sample identifiability artifact, not bias: the gating coefficient is centered at zero, but its empirical standard deviation across replicates (63.9) is roughly fifty times the median within-replicate Wald standard error (0.86). Forced to fit two classes to data with none, the optimizer latches onto random class boundaries and the Wald standard error does not absorb the labeling uncertainty. This is exactly the overextraction hazard predicted by Bauer and Curran (2003) and Lubke and Muthen (2005). The naive likelihood-ratio test is anti-conservative for a different reason, rejecting at 0.146, because the true reference distribution at the $ng = 1$ boundary is chi-bar-squared.

The BIC-conditional test controls size but is inert. Type I error 0.004 under the null, but power of only 0.025 at gating slope 1.0 and 0.004 at 1.5. At this N and DGP the BIC step selects a single class in essentially every replicate, so the procedure approximates a never-reject rule. Power is non-monotone in the gating slope, which the paper reads as the single-class LME absorbing the heterogeneous-slopes signal well enough that BIC never prefers two classes.

The standard analysis dominates. The linear-mixed `bm:Dbc` Wald test held nominal Type I (0 rejections of 240 under the null) and reached 0.542 power at gating slope 1.5, against 0.360 for the unconditional Wald and 0.004 for the BIC-conditional procedure. The unconditional test buys nothing for its size inflation: its power is below the LME baseline at both non-null cells.

A prior claim is corrected. The paper's own earlier proposition, that class-aware analyses are unusable as primary tests at small N, is narrowed. It holds for an unconditional Wald test on a single mixture coefficient, which requires a model-comparison wrapper (BIC or a bootstrapped LRT) to control size. The BIC-conditional procedure is size-valid; its problem is sensitivity, not validity.

What follows for practice

At $N \in [30, 150]$, a mixture model should not be the primary analysis for the biomarker-treatment interaction. Its legitimate roles are as a sensitivity check and as a DGP for stress-testing the fixed linear mixed model. If the simulation infrastructure is extended, the heterogeneous-random-slopes form is the right target, because it admits latent-class generative structure while leaving the analysis model a standard `nlme::lme` fit and preserving the audit chain.

The paper also warns that a sensitivity analysis run only under the covariance-only DGP will overstate carryover-induced power loss, since that specification puts the entire signal in the one channel carryover erodes by construction. Any signal flowing through the mean structure survives. Report power at three points on the spectrum: pure mean moderation, pure covariance moderation, and combined.

Scope and limitations

This is predominantly a theory and taxonomy paper with a pilot attached, and the paper says so. The five-study production program (approximately 2,300 ADEMP cells in the working protocol; estimated at about a week of background compute on 8-16 cores) has not been run, and the prazosin-PTSD re-analysis has not been attempted. The study is exploratory, and its findings should be confirmed in an independent study.

The empirical claims therefore rest on 240 replicates at a single sample size ($N = 70$), a single design, and one generative mechanism. The conclusion is a non-competitiveness result in that cell, not a general one: the regime where a class-aware test should win, large class separation with genuinely bimodal response, was never tested. Indeed the pilot's own DGP is one where the single-class model performs well, which is the paper's stated explanation for BIC's inertia and is a reason to expect the finding to be regime-specific. A canonical 1,000-replicate run at $N \in \{35, 70, 100\}$ over a wider gating-slope grid is required before the negative result can be generalized.

Two further cautions. The class-aware analysis costs roughly 32 times the linear-mixed fit per replicate, which is what confines the pilot to 240 replicates and rules out the bootstrapped LRT that would correctly calibrate the

likelihood-ratio test. And the reported Type I error of 0.000 for the LME baseline is a resolution artifact of 240 replicates rather than evidence of exact size; it bounds the rate below roughly 0.015, no tighter.

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